

Discussion of “Policymakers’ Uncertainty”

Ciéslak, Hansen, McMahon, Xiao

Andrea De Polis¹

¹Banco de España

Workshop on Advances in AI for Macroeconomic Analysis
Collegio Carlo Alberto

The views expressed in this presentation are those of the authors and do not necessarily reflect the views of the Banco de España or those of the Eurosystem.

The paper in one slide

Do policy makers care for uncertainty in their decisions?

The paper in one slide

Do policy makers care for uncertainty in their decisions? **Yes.**

The paper in one slide

Do policy makers care for uncertainty in their decisions? **Yes.**

- ① Individual FOMC members' risk perception → persistent balance of risks (BoRs)
 - LLM to extract BoR perceptions from FOMC statements
- ② Tilts in the BoR explain interest rate changes and unexpected surprises
 - R^2 of 6-10% for RR and AD monetary policy shocks
 - $P(\Delta i_{t+1} | BoR_t)$ increases (decreases) with inflation upside risks (gdp downside risks)
- ③ FOMC perceptions of the BoR predicts macro outcomes
 - Through policy actions and announcements

The paper in one slide

Do policy makers care for uncertainty in their decisions? **Yes.**

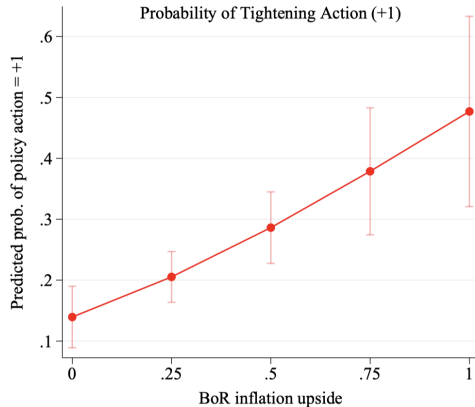
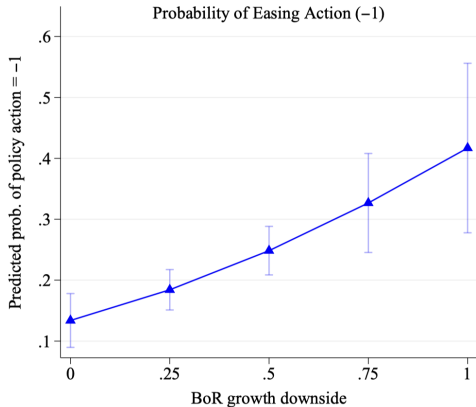
- ① Individual FOMC members' risk perception → persistent balance of risks (BoRs)
 - LLM to extract BoR perceptions from FOMC statements
- ② Tilts in the BoR explain interest rate changes and unexpected surprises
 - R^2 of 6-10% for RR and AD monetary policy shocks
 - $P(\Delta i_{t+1} | BoR_t)$ increases (decreases) with inflation upside risks (gdp downside risks)
- ③ FOMC perceptions of the BoR predicts macro outcomes
 - Through policy actions and announcements

Risk management approach to monetary policy



Insurance motives

Very telling results



As FOMC agreement about BoR increases, probability of expected policy action increases

Is insurance non-linear?

General loss-minimization FOC with endogenous tail risk:

$$2\bar{\Pi}'_t(r_t)(\bar{\Pi}_t(r_t) - \pi^*) + p'_t(r_t)(1 - 2p_t(r_t))\Delta_t^2 = -2\lambda\bar{Y}'_t(r_t)(\bar{Y}_t(r_t) - y^*)$$

Is insurance non-linear?

General loss-minimization FOC with endogenous tail risk:

$$2\bar{\Pi}'_t(r_t)(\bar{\Pi}_t(r_t) - \pi^*) + p'_t(r_t)(1 - 2p_t(r_t))\Delta_t^2 \overset{0}{=} -2\lambda\bar{Y}'_t(r_t)(\bar{Y}_t(r_t) - y^*)$$

- **Certainty equivalent:** no endogenous risk feedback

Is insurance non-linear?

General loss-minimization FOC with endogenous tail risk:

$$2\bar{\Pi}'_t(r_t)(\bar{\Pi}_t(r_t) - \pi^*) + p'_t(r_t)(1 - 2p_t(r_t))\Delta_t^2 = -2\lambda\bar{Y}'_t(r_t)(\bar{Y}_t(r_t) - y^*)$$

- Certainty equivalent: no endogenous risk feedback
- With insurance

$$\hat{r}_t = \underbrace{\frac{a}{a^2 + \lambda b^2}(\bar{\pi}_t - \pi^*) + \frac{\lambda b}{a^2 + \lambda b^2}(\bar{y}_t - y^*)}_{\text{Certainty equivalent}} + \underbrace{\frac{\Delta_t^2}{2(a^2 + \lambda b^2)}p'_t(r_t)(1 - 2p_t(r_t))}_{\text{Insurance}}$$

Is insurance non-linear?

General loss-minimization FOC with endogenous tail risk:

$$2\bar{\Pi}'_t(r_t)(\bar{\Pi}_t(r_t) - \pi^*) + p'_t(r_t)(1 - 2p_t(r_t))\Delta_t^2 = -2\lambda\bar{Y}'_t(r_t)(\bar{Y}_t(r_t) - y^*)$$

- Certainty equivalent: no endogenous risk feedback
- With insurance

$$\hat{r}_t = \underbrace{\frac{a}{a^2 + \lambda b^2}(\bar{\pi}_t - \pi^*) + \frac{\lambda b}{a^2 + \lambda b^2}(\bar{y}_t - y^*)}_{\text{Certainty equivalent}} + \underbrace{\frac{\Delta_t^2}{2(a^2 + \lambda b^2)}p'_t(r_t)(1 - 2p_t(r_t))}_{\text{Insurance}}$$

That is $\hat{r}_t = r_t^{CE} + \Omega(r_t) \rightarrow$ insurance is a linear shifter in a standard Taylor rule

Forward-looking expectations

Key assumption: *contemporaneous expectations*

$$\bar{\Pi}_t(r_t) = \bar{\pi}_t - ar_t \quad \text{and} \quad \bar{Y}_t(r_t) = \bar{y}_t - br_t$$

Forward-looking expectations

Key assumption: *contemporaneous expectations*

$$\bar{\Pi}_t(r_t) = \bar{\pi}_t - ar_t \quad \text{and} \quad \bar{Y}_t(r_t) = \bar{y}_t - br_t$$

Implications: clean and manageable *insurance wedge*

Forward-looking expectations

Key assumption: *contemporaneous expectations*

$$\bar{\Pi}_t(r_t) = \bar{\pi}_t - ar_t \quad \text{and} \quad \bar{Y}_t(r_t) = \bar{y}_t - br_t$$

Implications: clean and manageable *insurance wedge* , but...

No forward-looking constraints

- How does time-consistency square with the evidence on the tilts and future communication?
- Would $\hat{r}_t = r_t^{CE} + \Omega(r_t)$ still be optimal otherwise?
- Is this linear term enough?

The belief representation

Representation Theorem (De Polis et al. 2025)

To the first-order, a model with **(time-varying) asymmetric risks** can be represented as one with symmetric, zero-mean shocks, augmented by additional ***beliefs shocks***.

Cost-push shocks defined as a sequence of dummy surprise and anticipated shocks

$$u_t \equiv \sum_{j=0}^J \varphi_{t-j}^j.$$

The belief representation

Representation Theorem (De Polis et al. 2025)

To the first-order, a model with **(time-varying) asymmetric risks** can be represented as one with symmetric, zero-mean shocks, augmented by additional ***beliefs* shocks**.

Cost-push shocks defined as a sequence of dummy surprise and anticipated shocks

$$u_t \equiv \sum_{j=0}^J \varphi_{t-j}^j.$$

- ① **Dummy anticipated shocks** ensure that expectation revisions are consistent

$$\begin{aligned} \varphi_t^j &= E_t u_{t+j} - E_{t-1} u_{t+j} \\ &\approx \text{revisions in FOMC BoR perceptions} \end{aligned}$$

The belief representation

Representation Theorem (De Polis et al. 2025)

To the first-order, a model with **(time-varying) asymmetric risks** can be represented as one with symmetric, zero-mean shocks, augmented by additional ***beliefs* shocks**.

Cost-push shocks defined as a sequence of dummy surprise and anticipated shocks

$$u_t \equiv \sum_{j=0}^J \varphi_{t-j}^j.$$

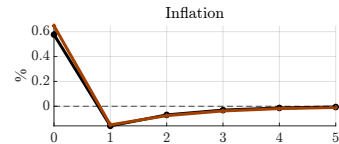
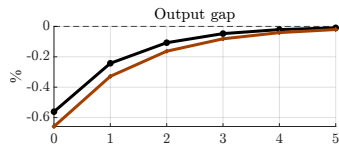
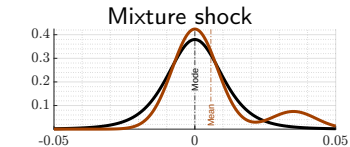
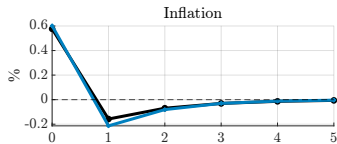
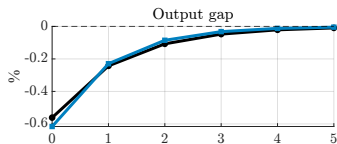
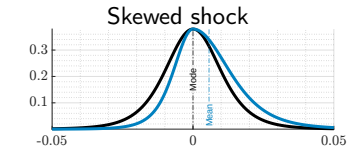
- ① **Dummy anticipated shocks** ensure that expectation revisions are consistent

$$\begin{aligned} \varphi_t^j &= E_t u_{t+j} - E_{t-1} u_{t+j} \\ &\approx \text{revisions in FOMC BoR perceptions} \end{aligned}$$

- ② **Dummy surprise shocks** φ_t^0 ensure that realized cost-push shocks are consistent

$$\varphi_t^0 = u_t - \sum_{j=1}^J \varphi_{t-j}^j$$

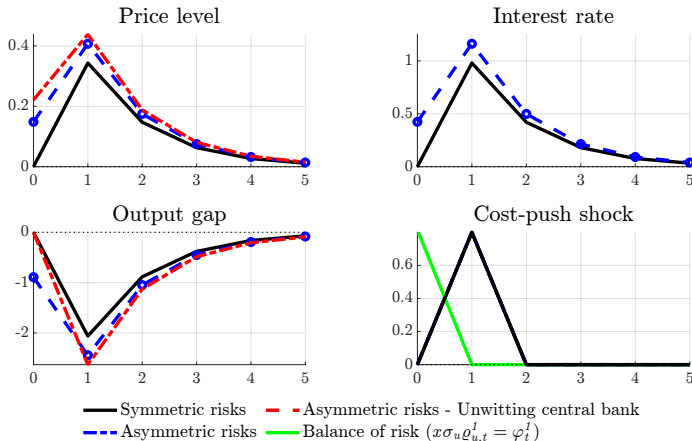
1) Linear is good enough



—●— Linear —●— Non-linear Skew-t —●— Non-linear Mixture with dominant mode

2) Linear is actually optimal

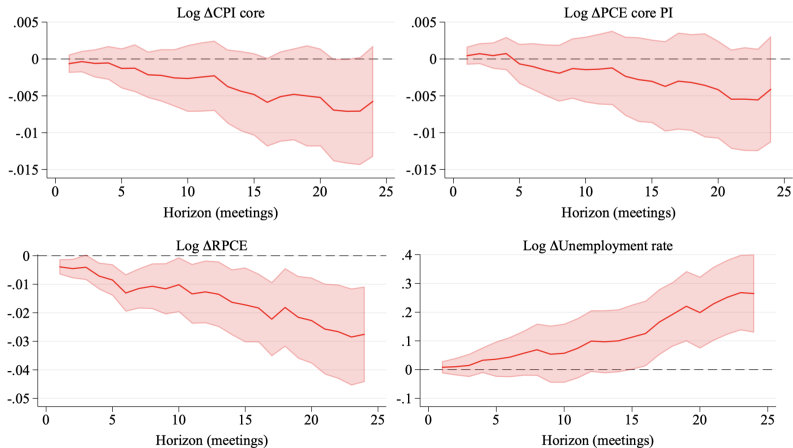
Within our setting $r_t = -(1 - \eta) \left[1 - \sigma \frac{k}{\alpha_x} \right] (\bar{p}_t + \lambda \varphi_t^1) \propto r_t^{CE} + \Omega(r_t)$



Lean against the balance of risks $\leftrightarrow$ insurance motives

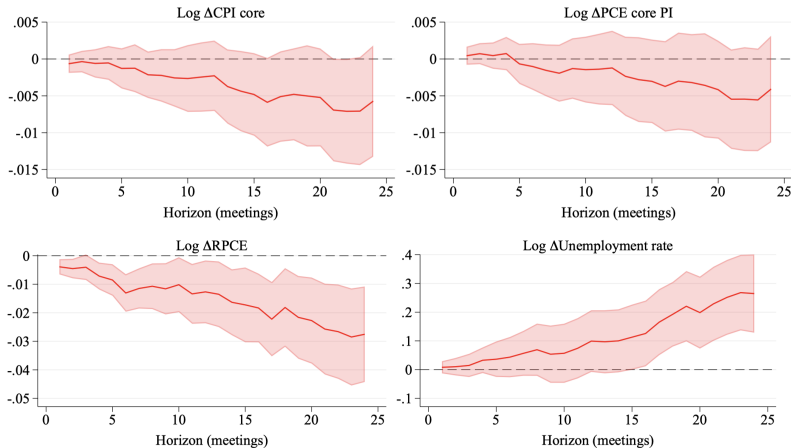
BoR, insurance and macro outcomes

$\Delta BoR_{t+h} > 0 \rightarrow$ insurance motives keep $\pi \approx 0$ at some real costs



BoR, insurance and macro outcomes

$\Delta BoR_{t+h} > 0 \rightarrow$ insurance motives keep $\pi \approx 0$ at some real costs

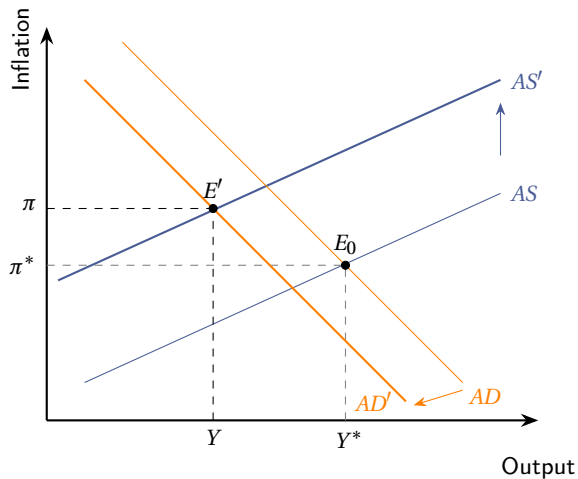


What is a BoR shock?

We document **cost-push dynamics** in response to a BoR shock

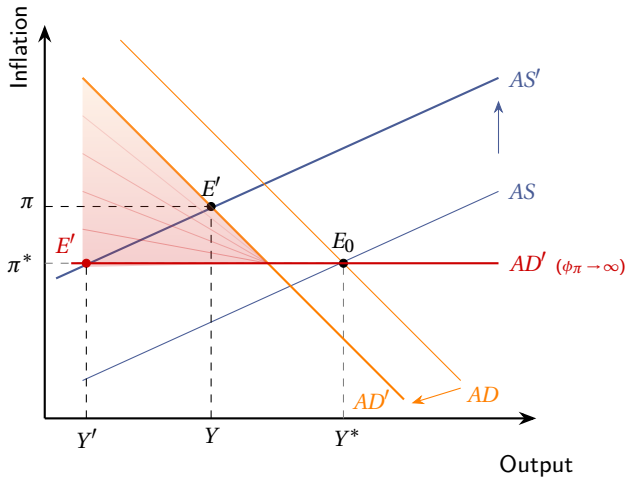
What about the policy instrument?

In the baseline NK model the slope of the AD = $-\frac{\sigma + \phi_y}{\phi_\pi}$



What about the policy instrument?

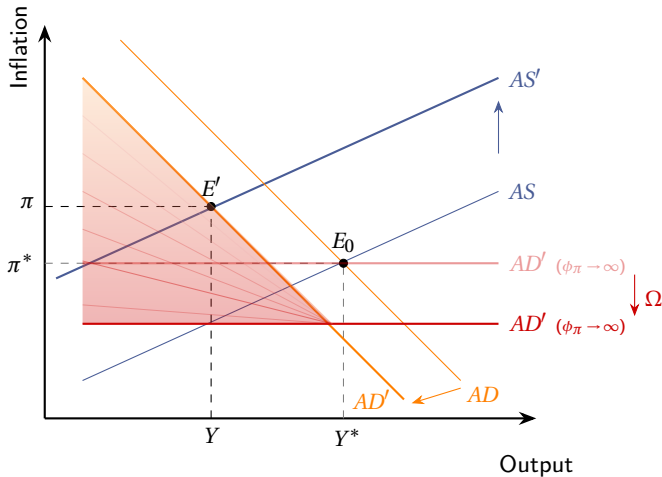
In the baseline NK model the slope of the AD = $-\frac{\sigma + \phi_y}{\phi_\pi}$



- ① Without insurance,
 $\phi_\pi \approx 10000$ to keep
 $\pi_t \approx 0$ on impact

What about the policy instrument?

In the baseline NK model the slope of the AD = $-\frac{\sigma+\phi_y}{\phi_\pi}$



- 1 Without insurance, $\phi_\pi \approx 10000$ to keep $\pi_t \approx 0$ on impact
- 2 Insurance Ω *only* shifts intercept: how big to keep ϕ_π to standard values?

Some further thoughts

for the coffee break

- ① How do the BoR results look for voters only?
 - Non-voters might have incentives to “tip the scale”
 - Do FOMC’s BoRs align with agents perceptions?
- ② Can the central bank really control p ?
 - To what extent can MP affect war-waging decisions? Or pandemic occurrences?
 - What if tail risks are independent? Or positively correlated?
- ③ In a world in which CBs communicate less, would insurance work the same?
 - How to conduct policy tilts?
 - How can we measure the subjective BoR then?
- ④ Could there be an attention threshold to BoR?
 - Agents pay more attention to inflation when it reaches 4% (Pfäuti 2026)
 - Is there a threshold for policy makers? Binscatters might reveal something already

Wrapping up

Very cool paper!

- Timely contributions to the literatures on monetary policy, communication, ...
 - BoR considerations as insurance policy
- State of the art measurement
 - BoR perceptions via LLM to analyze FOMC statements
- Bonus point: *two-way synergy* with our paper
 - Measured BoR and insurance motives meet NK framework with asymmetric risks

References

- De Polis, A., Melosi, L. & Petrella, I. (2025), The taming of the skew: Asymmetric inflation risk and monetary policy, Technical report, ECB Working Paper No. 3028.
- Pfäuti, O. (2026), 'The inflation attention threshold and inflation surges'.

Discussion of “Policymakers’ Uncertainty”

Ciéslak, Hansen, McMahon, Xiao

Andrea De Polis¹

¹Banco de España

Workshop on Advances in AI for Macroeconomic Analysis
Collegio Carlo Alberto

The views expressed in this presentation are those of the authors and do not necessarily reflect the views of the Banco de España or those of the Eurosystem.

The baseline nonlinear NK model

$$1 = \beta R_t^n E_t \left[\left(\frac{Y_{t+1}}{Y_t} \right)^{-\sigma} \frac{1}{\Pi_{t+1}} \right]$$

$$\psi \tilde{\Pi}_t \Pi_t = 1 - \epsilon_t + \epsilon_t w_t + \beta E_t \left[\Lambda_{t,t+1} \psi \tilde{\Pi}_{t+1} \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right]$$

$$R_t^n = (R_{t-1}^n)^{\rho_r} \left[\bar{R} \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \right]^{1-\rho_r}$$

Beliefs representation

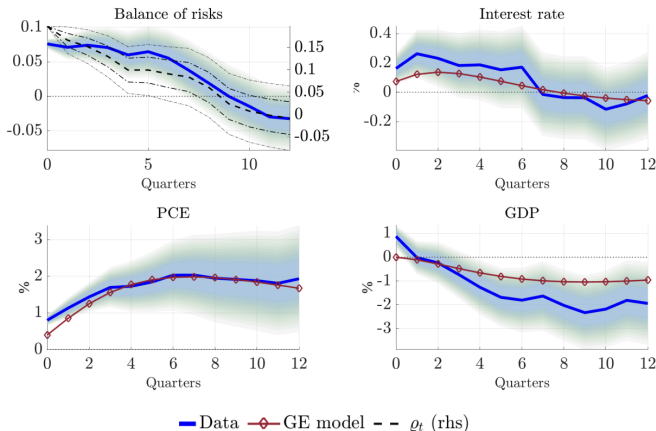
$$\hat{y}_t = E_t \hat{y}_{t+1} - \frac{1}{\sigma} (\hat{r}_t^n - E_t \hat{\pi}_{t+1})$$

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \kappa \hat{y}_t + \sum_{j=0}^J \varphi_{t-j}^j, \quad \varphi_t^j \sim \mathcal{N}(0, \sigma_j^2)$$

$$\hat{r}_t^n = \rho_r \hat{r}_{t-1}^n + (1 - \rho_r) (\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t)$$

What is a BoR shock? [◀ Back](#)

We use revisions in the BoR from a time-series model



Stagflationary dynamics: policy reaction more muted, inflation rises on impact.

- Could this reveal differences between subjective and “objective” BoR measurement?